Exploring the Sonic Universe of the Sega Genesis

Data Science, FM Synthesis, and 93,000+ Presets

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Ludo2026 — Session 9: Synthesis, Served Three Ways

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Who we are

Abraham Casas
(photo placeholder)

Abraham Casas

Electronics & CTO

Presenting today

Maria Cerro
(photo placeholder)

Maria Cerro

Design & CEO

Kasser Synths

A small studio exploring Yamaha FM chips and the sonic legacy of 1980s–90s games. **DAFMExplorer** is our open project for reading that legacy as structured data — and listening to it again.

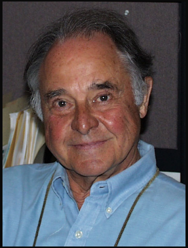
A platform's voice is also its presets

Beyond melodies and arrangements, Genesis music is built from **programmed FM timbres**:

- Reused across tracks and scenes
- Tuned for tiny TV speakers
- Authored under the constraints of one chip

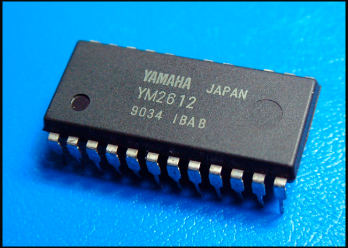
Those presets are creative decisions — and historical evidence.

From Chowning to the living room



Chowning / CCRMA

Yamaha DX7 (6 ops)

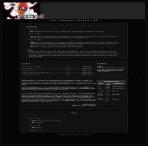


YM2612: **4 ops, 8 algorithms**

Same mathematical idea — frequency modulation — different creative solutions on every game.

The underexplored archive

- **VGM**: exact chip-command recordings
- **Project2612** / VGMrips: preservation at scale
- **DrWashington** collection: OPM presets extracted from those VGMs



93,000+ presets

A huge corpus of YM2612 voices — still underexplored as **structured data** in ludomusicology.

Live listening kit — games, music, presets

Four short live moments (VGM track → related preset). Keep each under ~20–25 s.

#	Game	Hear in the music	Then isolate a preset
L1	Streets of Rage	Koshiro's club energy	high-feedback lead / bass
L2	Pulseman	one-topology brand	Alg. 5 bright patch
L3	Nightmare Circus	GEMS-era US title	tool-shaped voice
L4	Thunder Force IV	dense shoot-'em-up FM	varied CON neighbour

VGM (offline)

Foobar2000 + in_vgm, or VGMPlay — files from VGMrips, cached on the laptop.

Preset (live)

DAFMExplorer filter by game → click → play YM2612.

https://dafm-explorer.vercel.app/

Data extraction — Making the archive speak

Turning raw OPM into an analysis-ready corpus:

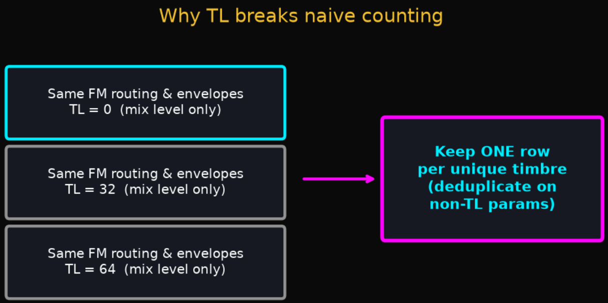
1. Parse presets (channel + 4 operators $\approx$ 58 parameters)
2. Deduplicate by timbre (ignore **TL** volume variants — same voice, different mix level)
3. Normalise game names (regional titles, aliases)
4. Enrich: composer, nationality, **GEMS** usage (Sega Retro, VGMrips, Wikipedia)

▶ VIDEO 1

From OPM file $\rightarrow$ a named preset with game, composer, and tool metadata

Placeholder — replace with *videos/V1.mp4*

Why TL breaks naive counting



Games reuse the same patch at different mix levels.

- **TL** = loudness in the arrangement, not a new timbre
- Counting every row would invent thousands of “fake” instruments
- Dedup on non-TL parameters keeps the creative decision once

What an FM preset is

Channel

- **CON** — algorithm (how operators connect)
- **FL** — feedback on operator 1
- Pan, LFO, ...

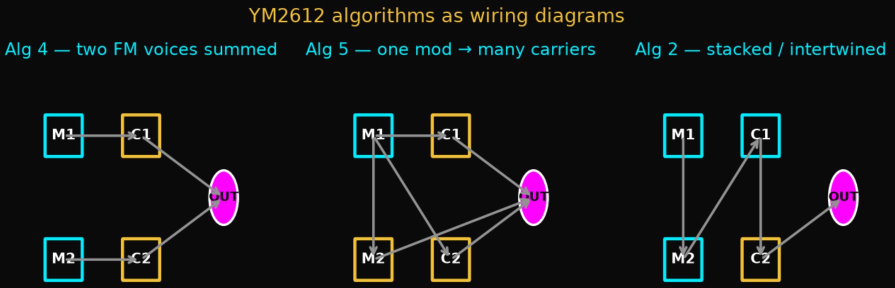
Four operators (M1, C1, M2, C2)

- Envelopes (AR, D1R, D2R, RR, ...)
- **TL** — level / mix
- **MUL** — frequency ratio

Carriers you hear; modulators shape their spectrum. Each preset = a creative solution under chip constraints.

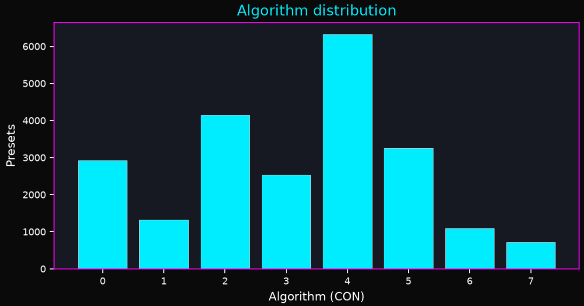
ｽﾛｯﾄ / ﾊﾟﾗﾒｰﾀ	CH1/CH4				CH2/CH5				CH3/CH6			
	S1	S3	S2	S4	S1	S3	S2	S4	S1	S3	S2	S4
DT/MULTI	30	34	38	3C	31	35	39	3D	32	36	3A	3E
TL	40	44	48	4C	41	45	49	4D	42	46	4A	4E
KS/AR	50	54	58	5C	51	55	59	5D	52	56	5A	5E
AM/DR	60	64	68	6C	61	65	69	6D	62	66	6A	6E
SR	70	74	78	7C	71	75	79	7D	72	76	7A	7E
SL/RR	80	84	88	8C	81	85	89	8D	82	86	8A	8E
SSG-EG	90	94	98	9C	91	95	99	9D	92	96	9A	9E
F-Num1	A0				A1				A2			
Block/F-Num2	A4				A5				A6			
‡F-Num1 ＊1									A9	A8	AA	A2
‡Block/F-Num2 ＊1									AD	AC	AE	A6
FB/Algorithm	B0				B1				B2			
L/R,AMS/PMS	B4				B5				B6			

Algorithms as wiring diagrams



Same four operators; different **CON** = different instrument architecture. Gold ≈ carriers, cyan ≈ modulators.

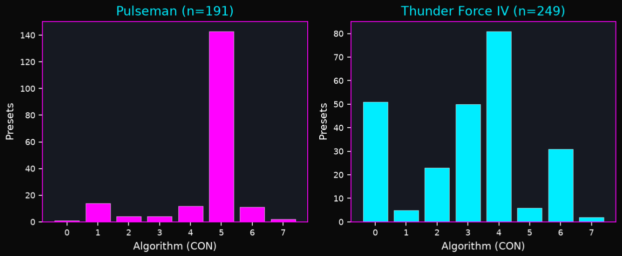
EDA: Algorithm 4 as the workhorse



- **Algorithm 4** dominates corpus-wide: two FM voices summed — flexible and mix-friendly
- Some scores commit hard to one topology
- Others spread routing across several algorithms

A shared “YM2612 grammar” — with room for fingerprints.

Game fingerprints — commit vs variety



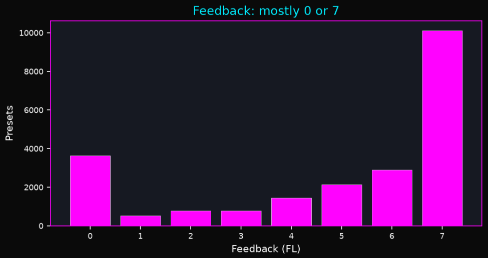
- **Pulseman**: one algorithm dominates (Alg. 5) — brand by commitment
- **Thunder Force IV**: spreads across CONs — identity from variety

Same chip, different house styles of sound design.

▶ **LIVE L2** Pulseman VGM → Alg. 5 preset in DAFMExplorer

videos/LIVE_DEMOS.md

EDA: Feedback as a binary aesthetic



Almost no middle ground: **FL = 0** or **FL = 7**.

- Max feedback: grit, cut, presence
- No feedback: clean, controllable

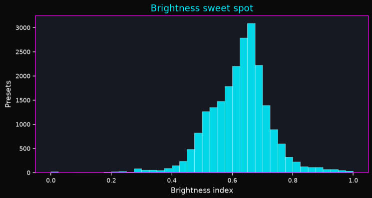
A stylistic switch, not a gentle trim.

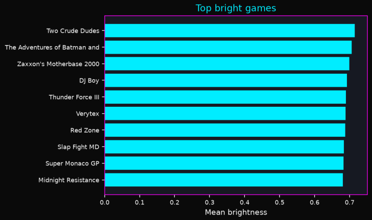
▶ VIDEO 2

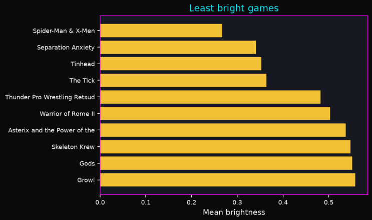
Hear FL 0 vs FL 7 (or two extreme presets)

Placeholder — replace with *videos/V2.mp4*

EDA: A brightness sweet spot





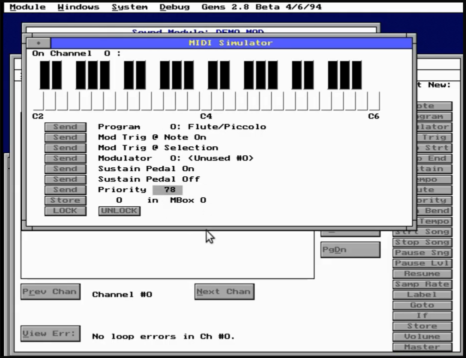


Brightness from TL, attack, and multipliers — presence on small TVs. Mid–high is the norm; a few titles stay deliberately darker.

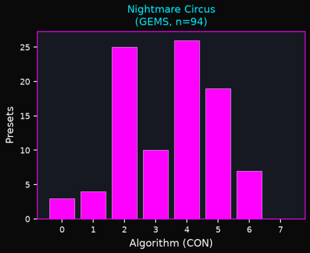
GEMS — the American sound driver

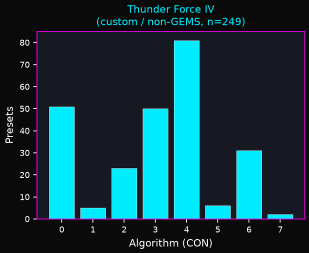
GEMS (Genesis Editor for Music and Sound): Sega of America's official tool (mid-1990s).

- Lowered the barrier to YM2612 programming
- Controversial: templates vs craft?
- In our corpus: a **tool variable**, not just a style label



GEMS in practice — a concrete contrast





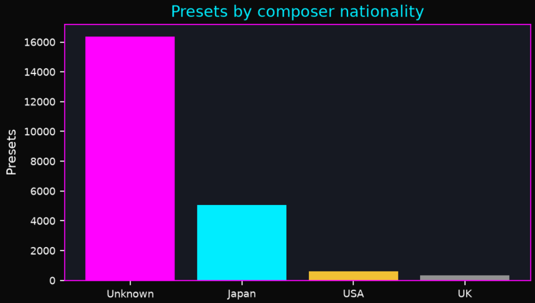
- **Nightmare Circus** (GEMS): shared tool ecosystem
- **Thunder Force IV** (non-GEMS): broader CON spread

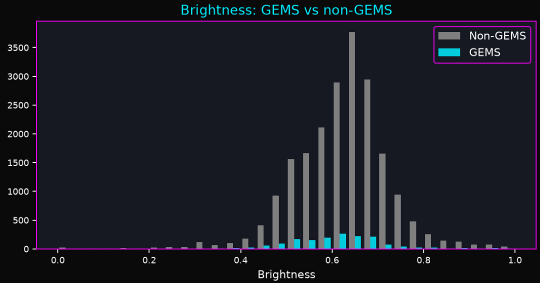
Not worse vs better — different industrial conditions.

▶ **LIVE L3+L4** Nightmare Circus vs Thunder Force IV: VGM + preset each

videos/LIVE_DEMOS.md

Composing by region / tool

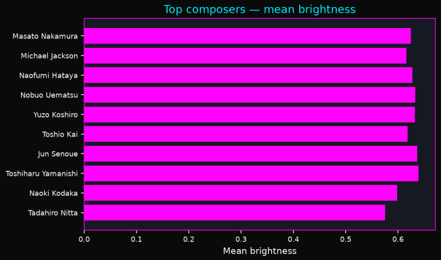




Japan / USA / UK as **hypotheses** about practice — drivers, deadlines, and shared templates shape timbre as much as “national style.”

Composer signatures

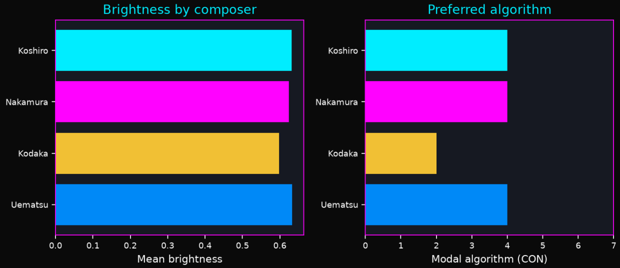
- **Yuzo Koshiro** — house / techno energy into YM2612 (*Streets of Rage*)
- **Masato Nakamura** — pop sensibility (*Sonic the Hedgehog*)
- Shared workhorse choices — individuality in envelopes, brightness, game fingerprints



▶ **LIVE L1** Streets of Rage VGM → high-FL Koshiro preset (VIDEO 3 only if live fails)

videos/LIVE_DEMOS.md

More signatures — Kodaka & Uematsu

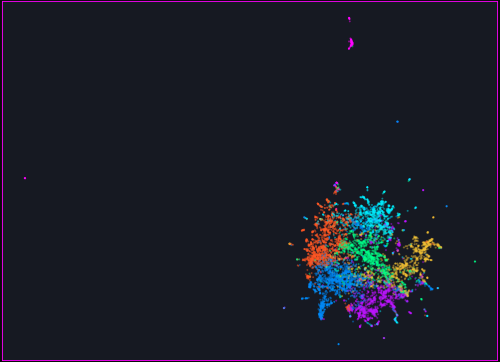


- **Naoki Kodaka**: modal CON leans away from the Alg. 4 default — more idiosyncratic routing
- **Nobuo Uematsu**: still near the workhorse grammar, with his own brightness / game palette

Convergence on tools of the chip $\neq$ identical sonic personalities.

Zooming out — timbral neighbourhoods

Timbral neighbourhoods (coloured by cluster)



At full corpus scale, presets form **neighbourhoods**: similar voices land near each other. Seven emergent "sonic realms" (from action neon to fantasy atmospheres) — a map of practice, not a taxonomy imposed in advance. No ML tutorial here — just the picture of the corpus.

What the data does not say

Honest limits keep the story scholarly:

- A preset is not a full score — melodies, sequencing, and PCM drums live elsewhere
- Extraction inherits archive quirks (naming, completeness, scraping gaps)
- Clusters and maps are **exploratory**: invitations to listen, not closed taxonomies
- “National style” and “GEMS sound” are hypotheses under industrial constraints

The payoff: better questions about platform practice — with sounds you can still hear.

Why this matters — and open tools

- Presets are evidence of **shared timbral practice** on a platform
- Extraction + exploratory analysis make composer, region, and tool differences discussable with data
- Method is open: code, documentation, and notebooks for reuse

Listen & explore

DAFMExplorer — interactive map, real-time YM2612 playback, filters.

`https://dafm-explorer.vercel.app/`

Next: same approach for other FM families (e.g. YM2151, YMF262).

Thanks — questions welcome

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https://www.kassersynths.com
https://dafm-explorer.vercel.app/
https://github.com/kassersynths/DAFMExplorer

Thank you — Q&A